Technical change and health: Evidence from the reduction of sugarcane burning in Brazil

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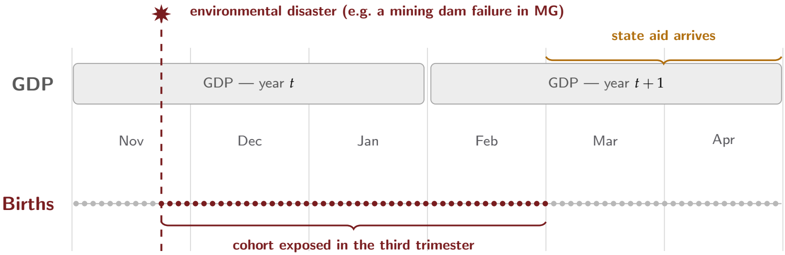
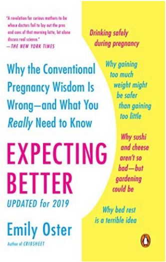
UFV

Health at birth: a thermometer that reads every day





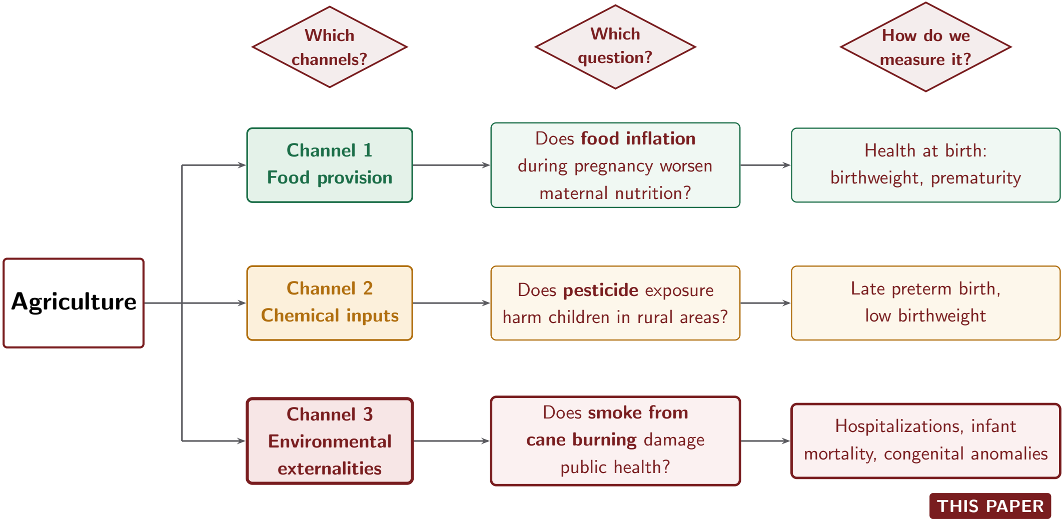
Emily Oster (Brown),

Expecting Better (2013)

- GDP is a great measure of development, but it arrives once a year
- Babies are born all the time — in 2024, **64** a day in Salvador, **46** in Curitiba, **2** even in Viçosa-MG

Birth records are high-frequency, precisely dated, and cover every birth – a thermometer of local conditions.

How does agriculture affect health?



Can a technical change in agriculture improve health?

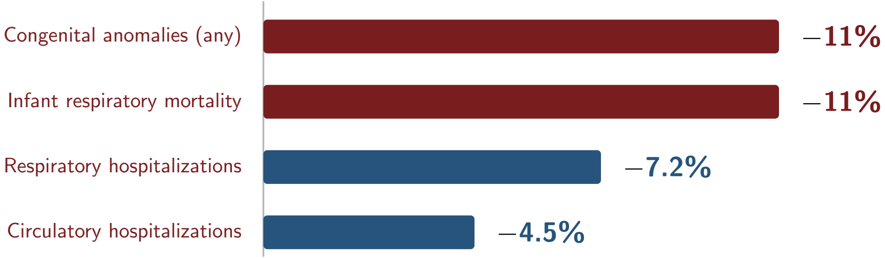
PRE-HARVEST BURNING ELIMINATED

2007: ≈33% mechanized → **2015**: >95% mechanized

Reduction per one-standard-deviation increase in exposure

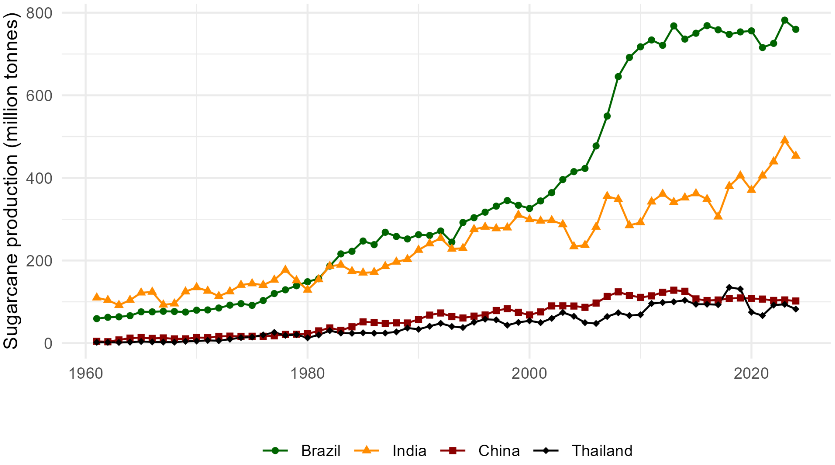
perinatal & infant

hospitalizations, all ages



And it really is the smoke: ✓ pollution fell **only** in harvest months ✓ **downwind** municipalities gained most

Context

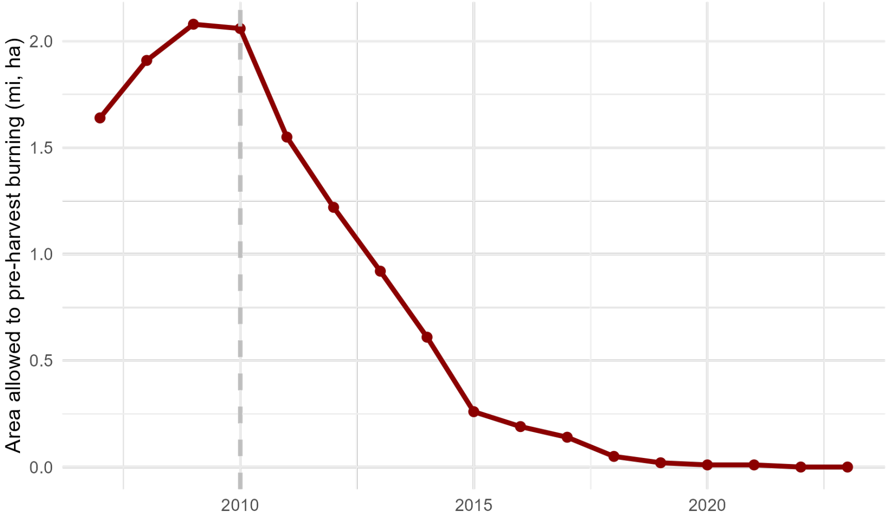

Sugarcane production (million tonnes)
800
600
400
200
0
1960
1980
2000
2020
Brazil
India
China
Thailand

Our Source of Identification

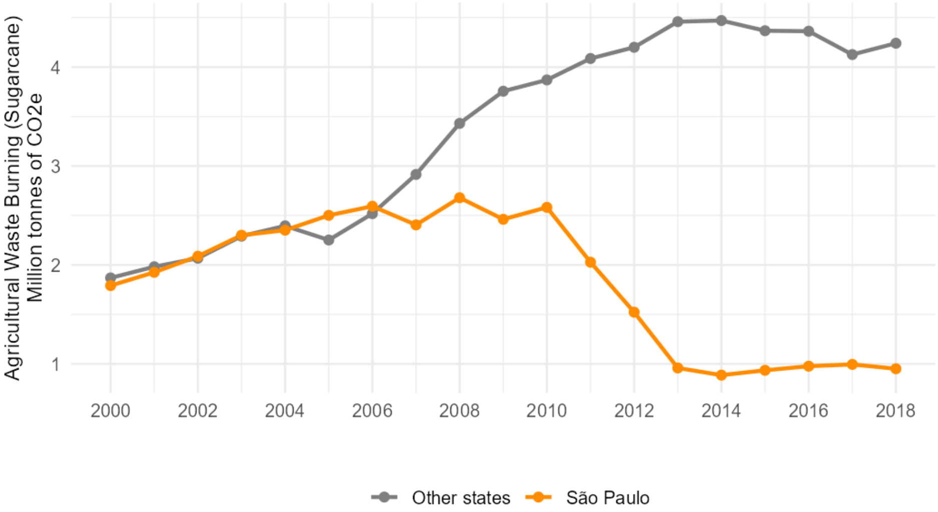
There was a reduction in burning driven by...

- Law
  - State Law 11.241/2002: gradual ban on straw burning (targets: 50% by 2011, 100% by 2021)
- Protocol
  - Green Ethanol Protocol (2007): voluntary industry agreement that accelerated deadlines to 2014–2017 in exchange for preferential financing and export market access
- Labor costs
  - Rising rural wages and export sustainability standards reinforced the shift [Davis, 2017]

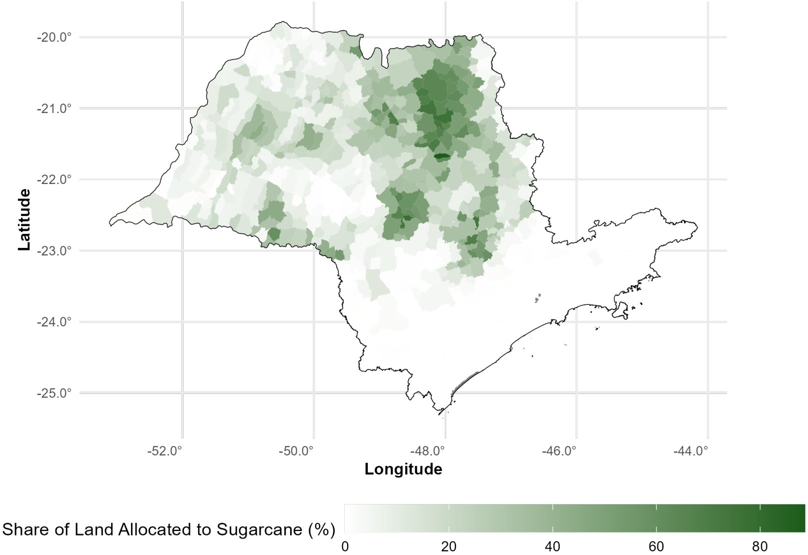
Area allowed to pre-harvest burning



Emissions from sugarcane agricultural waste burning



With spatial heterogeneity in this reduction...



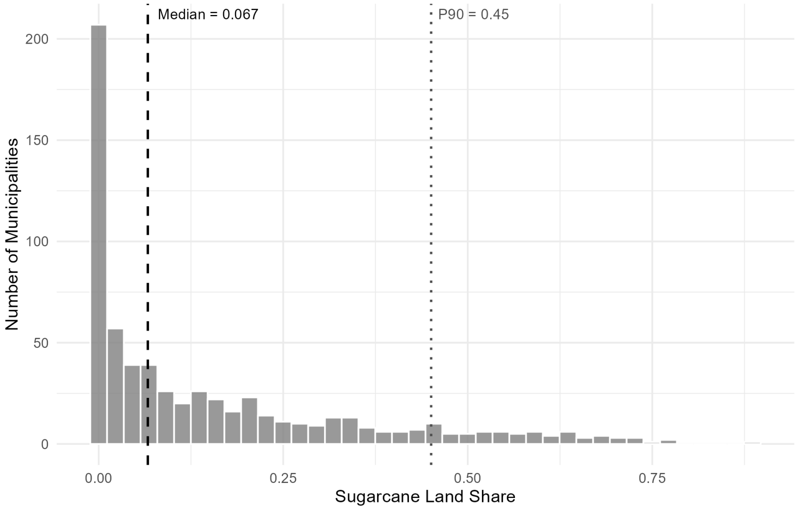
The Literature

- Rangel and Vogl [2019] – Review of Economics and Statistics: wind-based smoke exposure $\rightarrow$ lower birthweight, shorter gestation (São Paulo)
- Chagas et al. [2016] – Regional Science and Urban Economics: spatial DiD, sugarcane $\rightarrow$ respiratory hospitalizations, treatment is a binary median-split

Our contribution

- Broader outcomes: circulatory disease, infant mortality, a biologically-grounded congenital anomaly index
- Holding sugarcane production constant, what is the impact of reducing burning?
- Continuous rather than binary exposure, plus three sharper designs (season, wind, dose)

Heterogeneity matters



Combining the timing and the spatial heterogeneity

Difference-in-differences, continuous exposure

$$\ln\left(Y_{it}\right) \;=\; \alpha \cdot \underbrace{Land_i \times Post_t}_{\text{Tech. Change (TC)}} \;+\theta_i \;+\; \gamma_t \;+\; \varepsilon_{it}$$

$TC_{it} = Land_i \times Post_t$, with $Post_t = \mathbf{1}[t > 2010]$ and $Land_i =$ pre-2011 sugarcane land share

- θ_i, γ_t: municipality and year FE; regressions weighted by population (or births); SEs clustered by municipality
- $SD(Land_i) = 0.19$ — the unit for all "one-SD" effects below
- We also run specifications on individual birth records with mother fixed effects

Data Sources

- **Hospitalizations**: SIH/SUS, 2008–2022 (ICD J00–J99, I00–I99)
- **Births**: SINASC, 2007–2023, $N \approx 9.7$M individual records
- **Mortality**: SIM, 2002–2023
- **Treatment**: PAM-IBGE sugarcane land share, pre-2011 average
- **PM10/PM25**: CETESB monitoring network, 2003–2024
- **Fire**: MODIS MCD64A1 $\times$ MapBiomas LULC (cane pixels), Google Earth Engine
- **Wind**: NASA POWER/MERRA-2 reanalysis, Jul–Sep 2002–2022

All sources are administrative and cover the universe of events — no survey sampling error.

Results

Hospitalizations Fall for Respiratory & Circulatory Disease

$y = \ln(\text{hosp. per capita})$	Full (2008–2021)		Pre-Covid (2008–2019)	
	(1)	(2)	(3)	(4)
Panel A: Respiratory (J00–J99)				
Tech. Change	-0.380^{***}	-0.369^{***}	-0.287^{**}	-0.286^{**}
	(0.114)	(0.112)	(0.111)	(0.111)
Panel B: Circulatory (I00–I99)				
Tech. Change	-0.238^{***}	-0.224^{***}	-0.234^{***}	-0.222^{***}
	(0.078)	(0.075)	(0.078)	(0.075)
log(GDP/cap) control		X		X
Municipality & Year FE	X	X	X	X

* $p<0.1$, ** $p<0.05$, *** $p<0.01$. Weighted by population, clustered SEs.

- **1 SD more exposure** $\Rightarrow$ respiratory ↓**7.2%**, circulatory ↓**4.5%**; both stable when controlling for GDP/capita

Birth Outcomes: No Effect on Weight, Fewer Anomalies

	Any Anomaly (Q00–Q99)	Pol3 Index (NTD+Orofac.+Card.)	LBW (<2,500g)	VLBW (<1,500g)
Tech. Change	−0.006 66*	−0.004 27**	0.001 24	0.000 10
	(0.003 90)	(0.001 82)	(0.002 00)	(0.000 84)
Baseline mean	11.45/1000	3.59/1000	9.26%	1.47%

Individual (Sinasc), $N \approx 9.7$M, Controls: Mun., year, educ., sex, marital, race...

- **No effect on birthweight** (LBW, VLBW)
- **Congenital anomalies fall 11%**; the **Pol3 pollution-linked index falls 22.6%** at 1 SD of exposure
- Pol3 = neural tube defects + orofacial clefts + cardiac malformations

Which Anomalies?

$y = \Pr(\text{anomaly type})$	Neural Tube (Q00–Q07)	Orofacial (Q35–Q37)	Cardiac (Q20–Q26)	Pol3 Index
Tech. Change	$-0.000\,539^{**}$	$0.000\,000$	$-0.003\,846^{**}$	$-0.004\,265^{**}$
	$(0.000\,221)$	$(0.000\,156)$	$(0.001\,623)$	$(0.001\,823)$
Baseline (per 1,000)	1.09	0.73	1.91	3.59
Relative effect (1 SD)	-9.4%	≈ 0	$-\mathbf{38.3}\%$	-22.6%

Mortality: A Dose-Response Gradient

Adults — no effect

	Respiratory	Circulatory
Tech. Change	-0.028	-0.043
	(0.093)	(0.071)

$\Rightarrow$ -0.5% and -0.8% at 1 SD — statistically zero

Infants (<5) — significant

	Resp. mortality
Tech. Change	-0.593*
	(0.344)

$\Rightarrow$ -11% at 1 SD; all-cause infant mortality: null

Infants are physiologically more susceptible.

Is It Really the Smoke?

Less Fire and Less Particulate Matter

$$\ln\left(Y_{it}\right) = \alpha \cdot \underbrace{Land_i \times Post_t}_{\text{technical change – same treatment as before}} + \theta_i + \gamma_t + \varepsilon_{it}$$

$Y_{it} \in \{$ burned area on cane pixels, PM10, PM25 $\}$

	Harvest (Jul–Sep)	Off-season (Jan–Mar)
Panel A: Sugarcane burning (MODIS × MapBiomas)		
ln(burned area, cane pixels)	−4.964***	—
	(0.450)	
Panel B: Particulate matter (CETESB)		
ln(PM10)	−0.094***	−0.005
	(0.010)	(0.006)
ln(PM25)	−0.091***	−0.001
	(0.010)	(0.006)

*** p<0.01. Off-season burned area remains near zero throughout (placebo).

- Harvest-season burned area on cane fields falls ≈**66%** at mean exposure

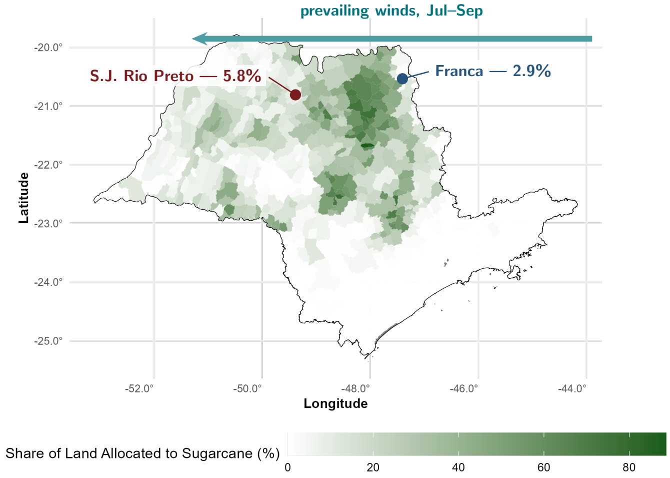
Health Gains Concentrate in Harvest Months

$$\ln Y_{itm} = \alpha \cdot TC_{it} + \beta \cdot (TC_{it} \times Harvest_m) + \theta_{i \times m} + \gamma_t + \varepsilon_{itm}$$

$y = \ln(\text{hosp./pop})$	Respiratory	Circulatory
Tech. Change (off-season)	-0.135^{***}	-0.083^{**}
	(0.033)	(0.028)
Tech. Change $\times$ Harvest	$\mathbf{-0.069}^{***}$	-0.014
	(0.011)	(0.013)

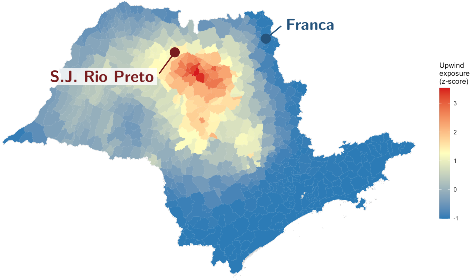
Respiratory hospitalizations fall more when burning would have occurred. Circulatory hospitalizations show no such difference.

Why the wind matters



$$WindExp_{it} = \sum_j \frac{\max\left(0, \cos\left(\hat{w}_{it} - \angle_{i \to j}\right)\right)}{dist_{ij}} \cdot SC_j$$

Smoke Travels with the Wind

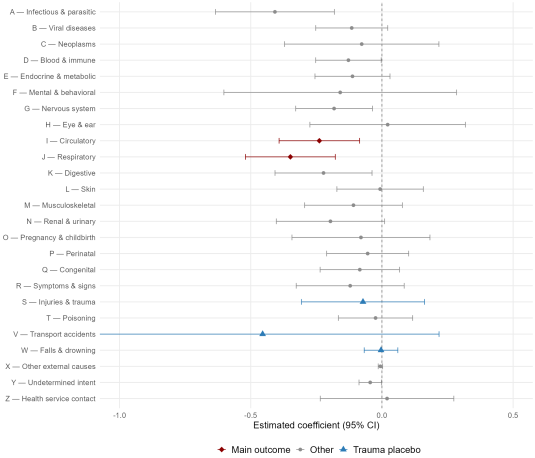


	Respiratory		Circulatory	
	Full	Pre-Cov.	Full	Pre-Cov.
Tech. Change	−0.348***	−0.265***	−0.238***	−0.222***
WindExp	−0.035*	−0.027	−0.033	−0.011

Municipalities downwind of cane fields — e.g. São José do Rio Preto — gain more than upwind ones such as Franca with identical own-exposure. Wind direction is plausibly exogenous to local health trends and, just as importantly, it is what the environmental economics literature now expects.

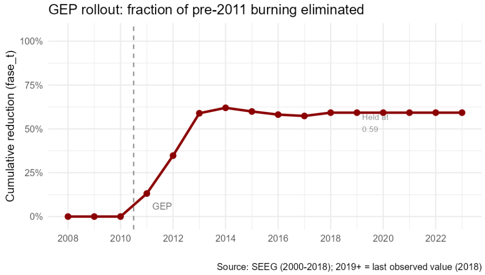
More Robustness

Placebo Across All ICD-10 Chapters



If the design were picking up general improvements in municipal health infrastructure, every chapter would move. It does not: only the two chapters with a biological link to particulate matter respond.

Continuous Treatment Intensity



	Respiratory	Circulatory
Binary (0/1)	-0.344^{***}	-0.239^{***}
Continuous ($\times fase_t$)	-0.618^{***}	-0.393^{**}

Replacing the post-2010 dummy with $fase_t$ — the fraction of burning emissions already eliminated — gives an implied effect at the mean of -0.346, essentially the binary estimate.

Robustness: Excluding São Paulo City

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
Baseline	-0.380^{***}	-0.287^{**}	-0.238^{***}	-0.234^{***}
Excl. SP city	-0.327^{**}	-0.236^{*}	-0.228^{**}	-0.189^{**}

São Paulo city has 12M inhabitants and zero cane share — a heavily weighted, mechanically untreated observation. Dropping it leaves the estimates essentially unchanged.

What Are the Stakes?

- Aggregating the hospitalization reduction over 2011–2021 across exposed municipalities: $\approx$**118,000** fewer hospital admissions (71,000 respiratory + 47,000 circulatory)
- At 2019 SUS reimbursement rates: **R$ 209 million** (USD 53M) in avoided hospitalization costs
- $\approx$**1,640** fewer births with cardiac malformations (2011–2023) $\Rightarrow$ **R$ 22 million** in avoided surgical costs

R$ 232 million (USD 59 million) in avoided costs over a decade — a conservative *lower bound* that excludes outpatient care, productivity losses, lifetime disability transfers, and the value of infant lives saved.

Takeaway

- A gradual, phased ban on pre-harvest sugarcane burning — backed by regulation *and* voluntary export-market incentives — delivered large, well-identified health gains
- The pattern across outcomes traces a **dose-response gradient**: null adult mortality $\rightarrow$ significant infant mortality $\rightarrow$ significant hospitalizations $\rightarrow$ significant congenital anomalies, each margin defined by a different vulnerability threshold
- Relevant beyond São Paulo: Northeast Brazil (Pernambuco: $>$90% manual), India, Thailand, Florida still rely on pre-harvest burning

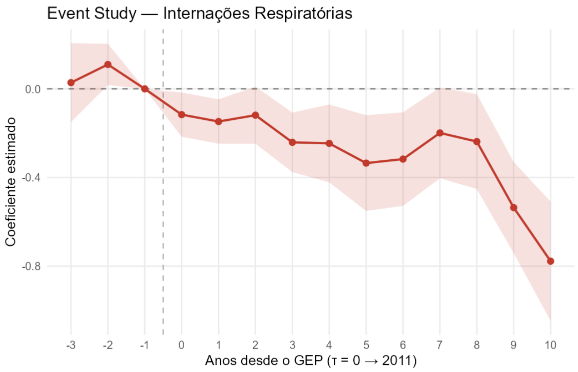
Thank you! Questions?

Backup Slides

Summary Statistics

Variable	N	Mean	Median	SD	Max
Sugarcane land share (2002–2010)	642	0.150	0.067	0.190	0.886
Respiratory hosp. per 1,000 pop.	8,385	7.281	—	4.939	45.828
Circulatory hosp. per 1,000 pop.	8,385	6.470	—	3.085	25.760
Any anomaly (per 1,000 births)	11,882	18.662	—	91.807	1,000
Infant resp. mortality (per 1,000 births)	9,029	0.702	—	3.204	111.111

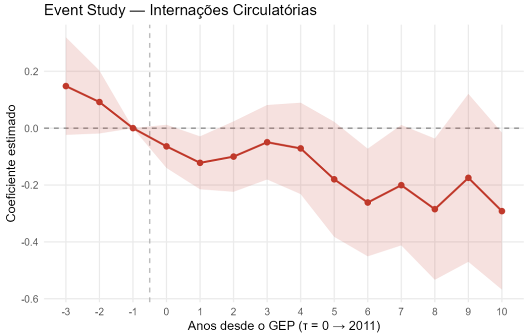
Identification: Parallel Trends Hold



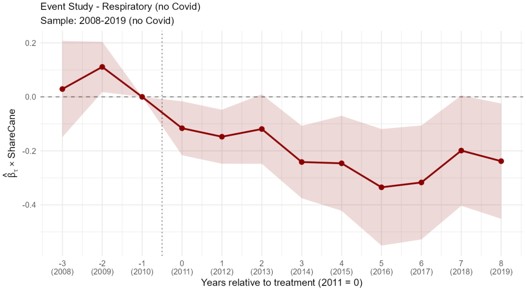
Pre-treatment coefficients ($\tau = -3, -2$) are near zero and insignificant; post-treatment effects grow steadily as the GEP phases in — consistent with a gradual, accelerating rollout, not a one-time break.

Event Study — Circulatory and Pre-Covid Samples

Circulatory (full sample)

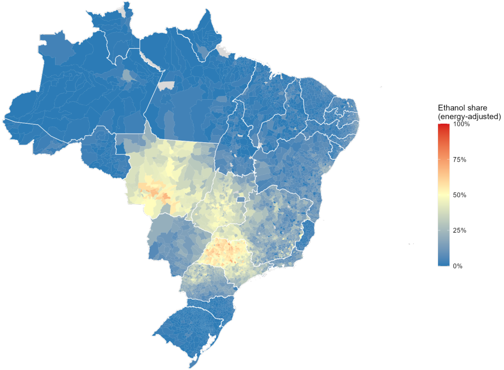


Respiratory (pre-Covid)



The same pattern survives dropping the pandemic years, so the result is not a Covid artifact.

Backup: Ethanol Market Context



Energy-adjusted hydrated ethanol share in the fuel mix, 2020–2024. Hydrated ethanol competes directly with gasoline where transport costs from the cane belt are low.

Backup: Aggregate Birth Outcomes (Municipality-Year)

	ln(birthweight)	ln(LBW)	ln(VLBW)	ln(anomalies)
Tech. Change	-0.002	0.027	-0.067	-0.795**
	(0.003)	(0.023)	(0.056)	(0.311)

The aggregate specification confirms the anomaly result; the individual-level LPM shown earlier additionally decomposes it by diagnostic group.

References I

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